

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

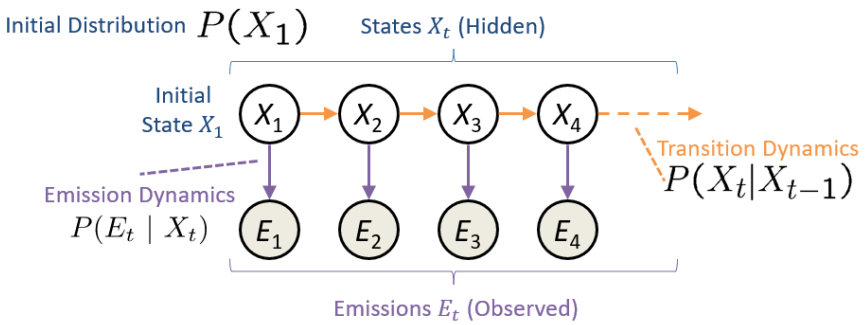
- Homework 2
  - Due **tomorrow 10/25 @5 pm**
- MOSS for code plagiarism checking
- Upcoming
  - Homework 3 and Classwork 5
- Slides will be uploaded to Brightspace before lecture.
  - Participation and interaction are still recommended.

This Lecture

- Filtering
  - Forward Algorithm
- Particle Filtering
  - Approximate Inference for HMMs

Last Lecture

- Hidden Markov Model (HMM)
  - Components
    - State X and Emission E
    - Emission Dynamics
    - Transition Dynamics
  - Assumptions
    - States are **not observed**
    - Emissions are **observed**



Last Lecture

- Inference Goal of HMM
  - $B(X_t) = P(X_t|e_{1:t})$
  - $B(X_t)$: Belief State
- Forward Algorithm (Filtering)
  - $B(X_t = x_t) = P(e_t|x_t) \sum_{x_{t-1}} P(x_t|x_{t-1}) \cdot \alpha B(x_{t-1})$

emission dynamics

transition dynamics

belief of being in last state

Forward Algorithm as Dynamic Programming

- Computing $B(X_t)$ follows **repeated** application of **two steps** at all previous time steps:
  - Time Update

$$B'(x_t) = P(x_t|e_{1:t-1}) = \sum_{x_{t-1}} P(x_t|x_{t-1}) \cdot B(x_{t-1})$$

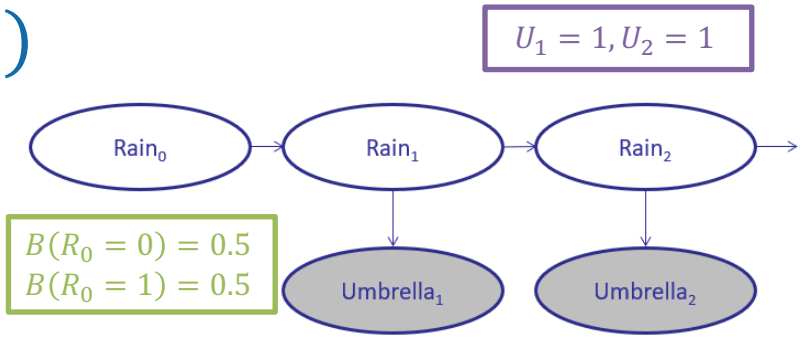
  - Observation Update

$$B(x_t) = P(x_t|e_{1:t}) \propto P(e_t|x_t)B'(x_t)$$

Example

- Find $P(R_2 = 1 | U_1 = 1, U_2 = 1)$
- **Step 1 – Setup and Priors**

	t=0	t=1	t=2
Time Update $B'(R_t)$	---		
Observation Update $B(R_t)$	$B(R_0 = 0) = 0.5$ $B(R_0 = 1) = 0.5$		



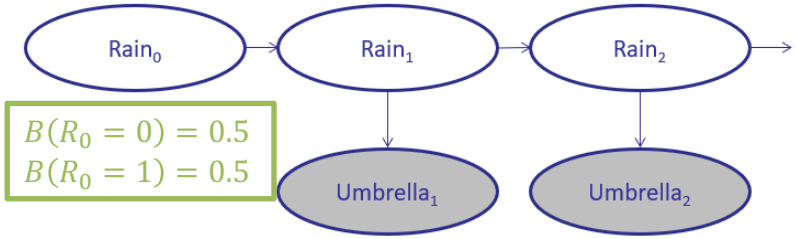
R_t	R_{t+1}	$P(R_{t+1} \mid R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

Transition Dynamics

R_t	U_t	$P(U_t \mid R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

Emission Dynamics

Example



R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t\|R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

- **Step 2 – Time Update**

$$B'(x_t) = \sum_{x_{t-1}} P(x_t | x_{t-1}) \cdot B(x_{t-1})$$

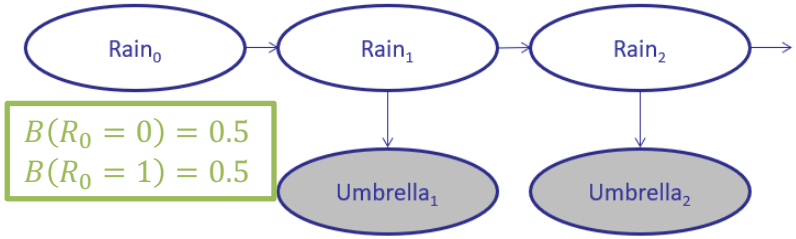
$$B'(R_1 = 0) = \sum_{r_0} P(R_1 = 0 | R_0 = r_0) \cdot B(R_0 = r_0)$$

$$= P(R_1 = 0 | R_0 = 0) \cdot B(R_0 = 0) + P(R_1 = 0 | R_0 = 1) \cdot B(R_0 = 1)$$

$$= 0.7 * 0.5 + 0.3 * 0.5 = 0.5$$

	t=0	t=1	t=2
Time Update $B'(R_t)$	---	$B'(R_1 = 0) = 0.5$ $B'(R_1 = 1) = 0.5$	
Observation Update $B(R_t)$	$B(R_0 = 0) = 0.5$ $B(R_0 = 1) = 0.5$		

Example



$B(R_0 = 0) = 0.5$
$B(R_0 = 1) = 0.5$

$U_1 = 1, U_2 = 1$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t\|R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

- **Step 3 – Observation Update**

$$B(x_t) \propto P(e_t|x_t)B'(x_t)$$

$$B(R_1 = 0) \propto P(U_1 = 1|R_1 = 0)B'(R_1 = 0)$$

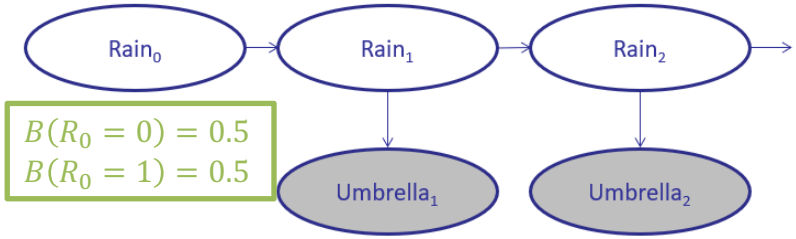
$$\propto 0.2 * 0.5 = 0.1 \Rightarrow {}^{0.1}/_{0.1 + 0.45} = 0.182$$

$$B(R_1 = 1) \propto P(U_1 = 1|R_1 = 1)B'(R_1 = 1)$$

$$\propto 0.9 * 0.5 = 0.45 \Rightarrow {}^{0.1}/_{0.1 + 0.45} = 0.818$$

	t=0	t=1	t=2
Time Update $B'(R_t)$	---	$B'(R_1 = 0) = 0.5$ $B'(R_1 = 1) = 0.5$	
Observation Update $B(R_t)$	$B(R_0 = 0) = 0.5$ $B(R_0 = 1) = 0.5$	$B(R_1 = 0) = 0.182$ $B(R_1 = 1) = 0.818$	

Example



- **Step 4 – Repeat Steps 2-3**
  - Step 2: Time Update

$$B'(x_t) = \sum_{x_{t-1}} P(x_t | x_{t-1}) \cdot B(x_{t-1})$$

  - Step 3: Observation Update

$$B(x_t) \propto P(e_t|x_t)B'(x_t)$$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t\|R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

	t=0	t=1	t=2
Time Update $B'(R_t)$	---	$B'(R_1 = 0) = 0.5$ $B'(R_1 = 1) = 0.5$	$B'(R_2 = 0) = 0.373$ $B'(R_2 = 1) = 0.627$
Observation Update $B(R_t)$	$B(R_0 = 0) = 0.5$ $B(R_0 = 1) = 0.5$	$B(R_1 = 0) = 0.182$ $B(R_1 = 1) = 0.818$	$B(R_1 = 0) = 0.117$ $B(R_1 = 1) = 0.883$

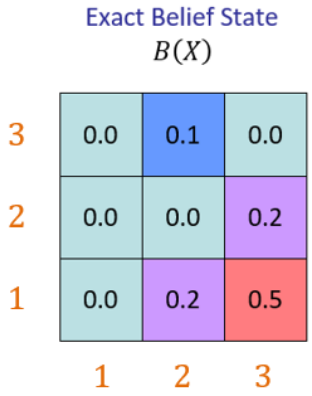
Exact Inference vs. Approximate Inference

- What are the differences?

		Exact Inference	Approximate Inference
Differences		• Mathematically precise, guaranteed correct results • Computationally expensive	• Estimated results • Trades accuracy for computational efficiency
Approach	BN	• Exact Inference in O(1), $O(\|V\|)$, $O(\|V\|^2)$ • Enumeration Inference	• Prior Sampling/Rejection Sampling • Gibbs Sampling
	HMM	• Forward Algorithm	• Particle Filtering

Particle Filtering

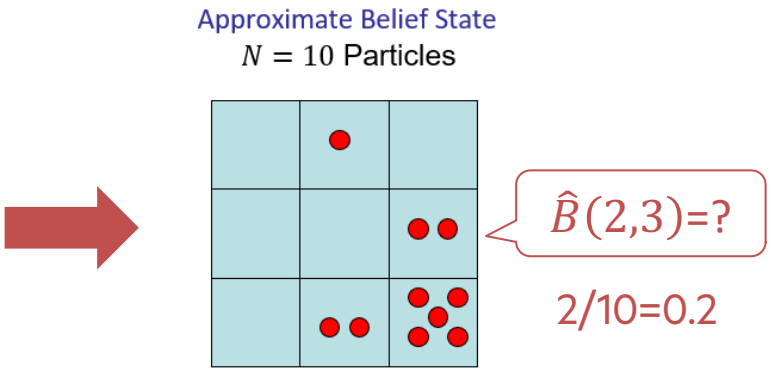
- Representing the belief state $B(X)$ as **a list of N particles**, which are samples of $B(X)$ that aggregate around the most likely states.



State $X = (row, col)$:
location of ghost

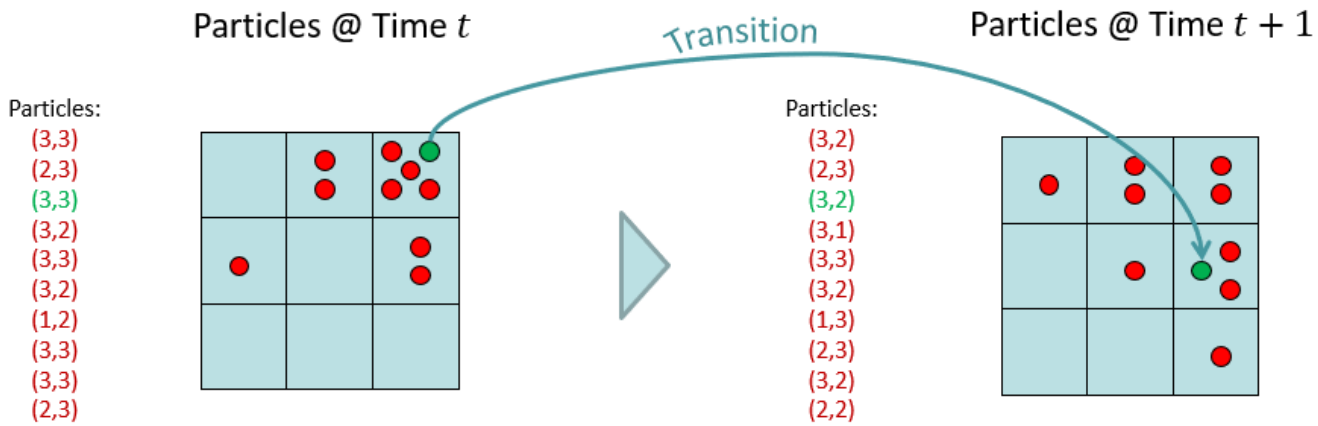
Transition $P(X_t|X_{t-1})$:
movement of ghost

Emission $P(E_t|X_t)$:
sensor reading



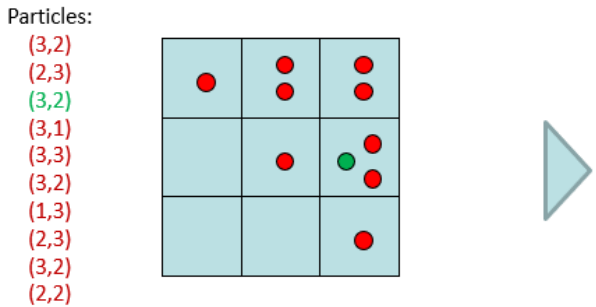
Particle Filtering

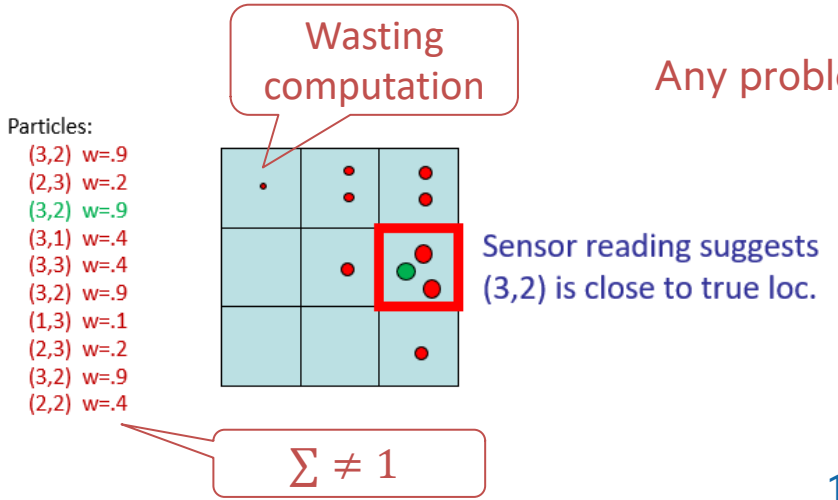
- **Step 1 – Elapse Time**
  - Sample the next state of each particle from transition $P(X_{t+1}|X_t)$
  - $x_{t+1} = sample(P(X_{t+1}|X_t))$



Particle Filtering

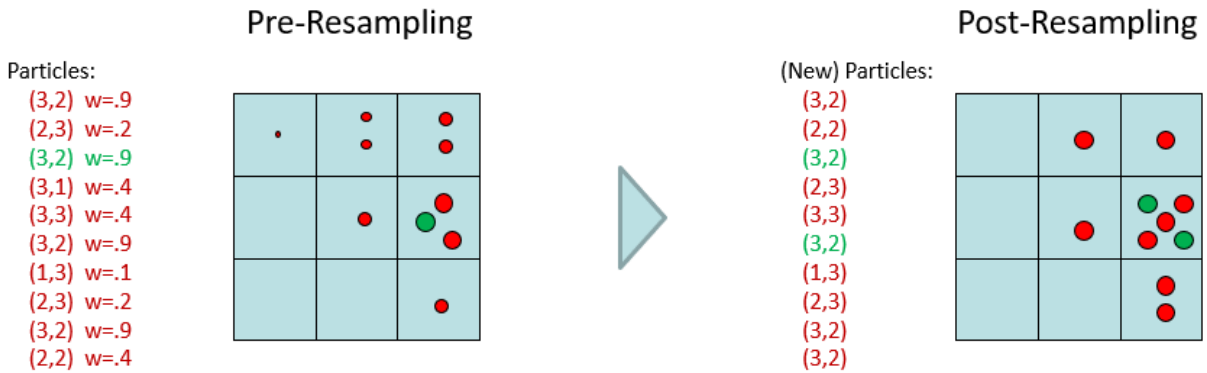
- **Step 2 – Observing Evidence e**
  - **Downweight** each sample based on the **emission dynamics**
  - For each particle x, assign some **weight** $w(x_t) = P(e_t|x_t)$
  - $B(X) = P(e_t|x_t) * B'(X)$





Particle Filtering

- **Step 3 – Resampling/Normalization**
  - For all N particles in the filter, **sample** a new particle (with replacement) from the weighted distribution found in Step 2.
  - New particles are now equally weighted!

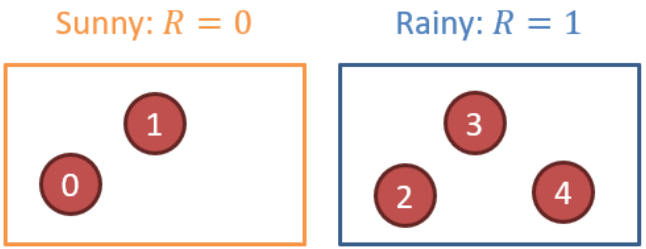


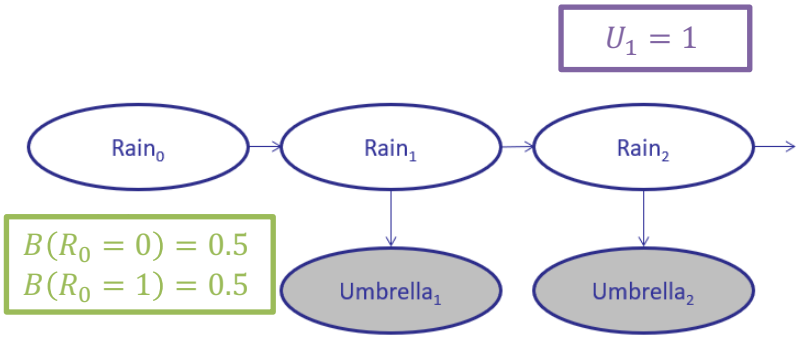
Particle Filtering

- **Step 4: Repeat**
  - Elapse Time
  - Observing Evidence e
  - Resampling

Example

- Use N=5 particles to estimate $B(X_1)$, and observe $U_1 = 1$



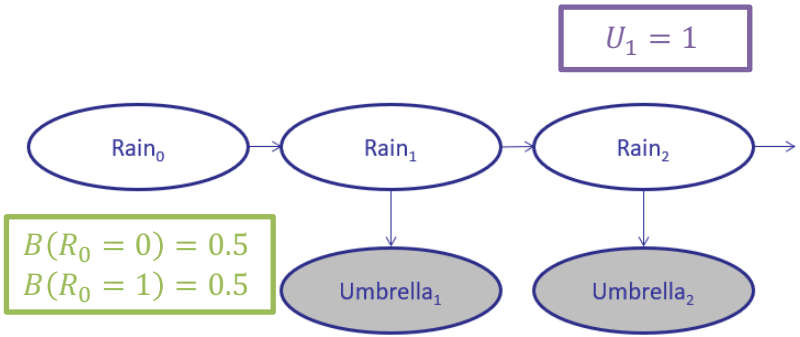


R_t	R_{t+1}	$P(R_{t+1} \mid R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t \mid R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

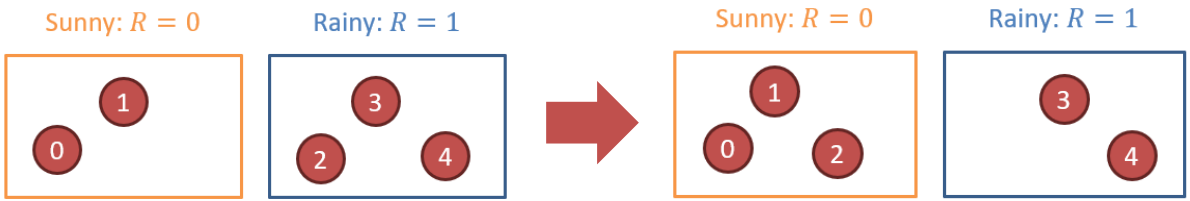
Example

- Step 1: Elapse Time
  - E.g., what the likelihood that particle 0 stays in state R=0?
    - $P(R_{t+1} = 0 | R_t = 0)$



R_t	R_{t+1}	$P(R_{t+1} \mid R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

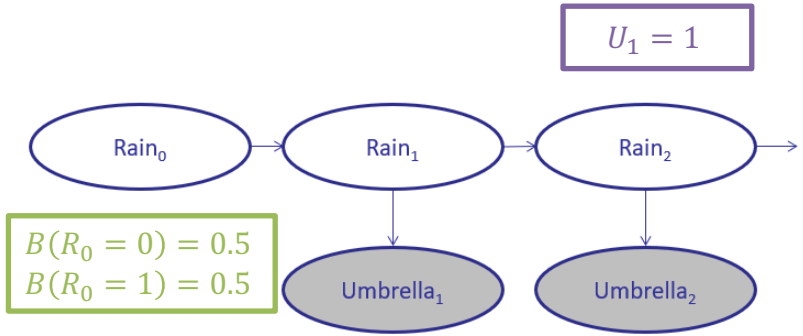
R_t	U_t	$P(U_t \mid R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9



A possible state of the particles

Example

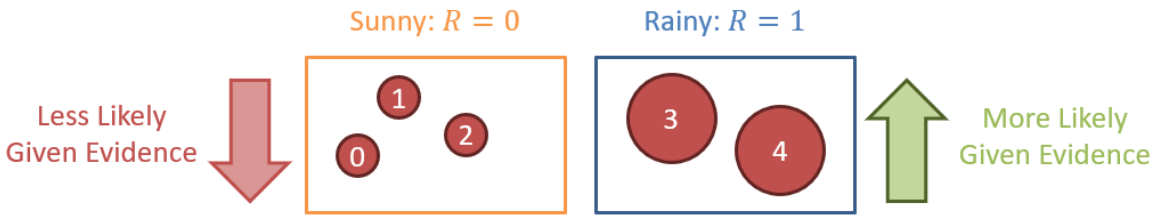
- Step 2: Observing Evidence
  - $U_1 = 1$
  - Reweight $w(R) = P(U_1 = 1|R)$

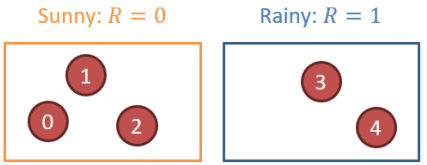


R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t\|R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

Particles	0	1	2	3	4
$w(x)$	$P(U = 1\|R = 0)$ 0.2	$P(U = 1\|R = 0)$ 0.2	$P(U = 1\|R = 0)$ 0.2	$P(U = 1\|R = 1)$ 0.9	$P(U = 1\|R = 1)$ 0.9

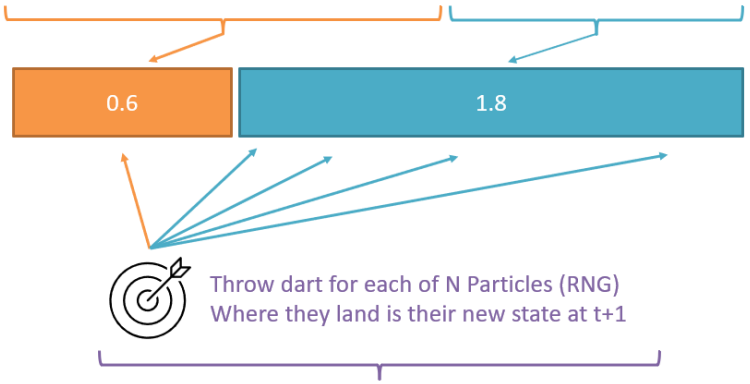




Example

- Step 3: Resampling/Normalization

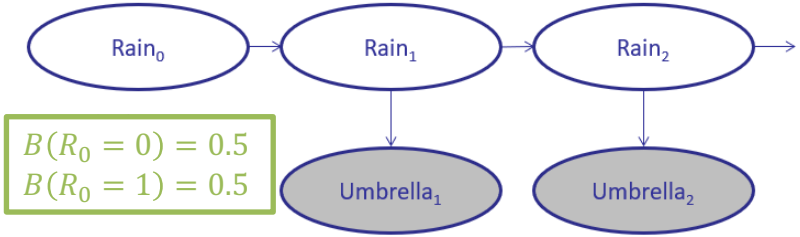
Particles	0	1	2	3	4
$w(x)$	$P(U=1\|R=0)$ 0.2	$P(U=1\|R=0)$ 0.2	$P(U=1\|R=0)$ 0.2	$P(U=1\|R=1)$ 0.9	$P(U=1\|R=1)$ 0.9



Particles	0	1	2	3	4
X_{t+1}	$R=0$	$R=1$	$R=1$	$R=1$	$R=1$



$U_1 = 1$



R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
0	0	0.7
0	1	0.3
1	0	0.3
1	1	0.7

R_t	U_t	$P(U_t\|R_t)$
0	0	0.8
0	1	0.2
1	0	0.1
1	1	0.9

$$B(R_1=0)=\frac{1}{5}=0.2$$

$$B(R_1=1)=\frac{4}{5}=0.8$$